Fault tolerance and recovery

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Sample chapter for the proposed course plan.

1 Learning outcome

Describe a recovery plan for a crash during request processing.

Prerequisites: Lectures 03 and 07.

2 Core idea

Recovery depends on knowing which state survives a failure. A process restart removes volatile state but may preserve durable records. A correct recovery path must reconcile those records with requests that clients may retry.

3 Worked scenario

A worker records that a registration task finished, then crashes before acknowledging the queue message. The task is delivered again.

4 Discussion

How can the worker recognize that the repeated task was already completed?

5 Recap

Design recovery and normal processing together.